

Discrete Optimization

Production and inter-facility transportation
scheduling for a process industryRenato de Matta ^{a,*}, Tan Miller ^b^a *Department of Management Sciences, College of Business Administration, University of Iowa,
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Received 21 May 2002; accepted 22 December 2002

Available online 5 August 2003

Abstract

We study in this paper the problem of coordinating the short-term production and inter facility transportation scheduling decisions between a plant that produces intermediate products and a finishing plant which processes the intermediate products into finished goods. Our goal is to develop a better understanding of the general relationships between production and transportation scheduling decisions, in particular, how changes in plant capacity and costs affect the coordination of scheduling decisions as well as the choice of transportation modes and carriers. We formulate the problem as a mixed integer programming model and use a solution procedure that exploits the underlying structure of our problem. Using real and simulated data from a process industry firm, our computational study, which compares the production and transportation schedules obtained from coordinated scheduling and sequential scheduling, shows that coordinated schedules yield significant cost savings resulting from the modest use of the expensive fast transport mode, coordinated product changeovers between plants and reduced intermediate product inventories. Discussions of managerial implications of coordinated scheduling are presented.

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Keywords: Production planning and scheduling; Transportation scheduling; Inventory/production; Deterministic

1. Introduction

Achieving high equipment utilization is a major concern among process industries due to the large capital investments on machines and equipment. Process industries generally use a capacity-oriented planning and scheduling (COPS) framework which, in contrast to fabrication and assembly industries, schedules capacity before raw materials. COPS consists of resource requirements planning, production planning, and operational planning and scheduling (Taylor et al., 1981). Briefly, the resource requirements planning develops the long range strategic plans to acquire the resources necessary for future operations.

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Production planning addresses the utilization of those resources over an intermediate range, i.e., annual or quarterly. Finally, operational planning determines the short range detailed specifications of what is required, when it should be produced, and where it is produced.

Of particular interest in this paper is the coordination of the production scheduling of finished products and intermediate products in the operational planning level. Coordination represents a significant operational challenge when the production facilities (plants) that produce the intermediate products and finished products are geographically separated because the timing of inter facility transfers of intermediate products between plants also requires attention. Process industry firms with multi-echelon manufacturing networks must decide when to ship the intermediate products, how much to ship and how to select among several transport modes with differing costs and transit times.

Frequently, process industry firms choose not to schedule the production facilities and their transportation links in a simultaneous, integrated fashion. Instead, they tend to schedule them either independently or sequentially by letting the production schedule at the finishing plant drive the production requirements of its supplying intermediate plant. For example, a “mix and pack” manufacturer will blend ingredients to produce batches of intermediate products which are stored and piped to automated packaging equipment. Oftentimes, packaging is the bottleneck operation and the scheduling of the entire mix and pack process is oriented around it, i.e. the production, accumulation and flow of the intermediate products are scheduled as required based on the production schedule, inventory levels and inventory requirements of the finished products.

If the individual processes are located at different geographic sites, the lack of coordination in scheduling the production operations could substantially suboptimize a firm’s production and transportation costs and potentially reduce its operating efficiency. Consider for instance the computer industry. Our experience working in this industry shows some firms utilize air carriers for a significant portion of their transport volume between plants, and also between plants and distribution centers. Firms in this industry and other process industries typically do, however, have the option of employing multiple transport modes, including cheaper surface transport, for most of their shipments (i.e., the materials being transported generally do not require that air rather than surface transport be used). In some cases, air transport may represent the best choice. Our experience indicates that often even in situations where cheaper surface transport represents the most economical choice (when all logistics costs are considered), some firms often still use air shipments. We have observed that air shipments frequently occur because of intermediate product inventory shortages at a finishing plant which, in most cases, result from the lack of coordinated scheduling between plants.

We present in this study an integrated approach to coordinate and synchronize the production and transportation scheduling decisions of plants that are linked by transport lanes and carriers. The paper is organized as follows. Section 2 describes the coordination problem in detail. Section 3 offers a brief review of the literature on related scheduling problems. In Section 4, we describe and formulate our production and transportation scheduling model. Section 5 details our approach to solve the proposed model. Section 6 presents the numerical results for a large variety of test problems. The managerial implications of coordinated production and transportation scheduling are presented in Section 7. Finally in Section 8, we offer some concluding remarks about our study.

2. Problem description

Consider a process industry that has two plants A and B. Plant A produces several families of intermediate products which are shipped directly to B in varying amounts for processing into families of finished products. Each individual plant has several production lines where every production run on each line, which could be for a duration of one or several consecutive weeks, can produce only one product family at

a time. Using an “all or nothing” production policy, a production line produces to capacity in any period (see Magnanti and Vachani, 1990) or a fixed quantity (that is less than capacity) which is specified in the production plan (Miller, 2002). Nonsequence-dependent changeover costs and/or time are incurred at each plant whenever a production line changes production to a different product family. Additionally, two general transport mode categories are employed in direct shipments of intermediate products from A to B, i.e., a normal and an expedited transport mode. The expedited or faster mode consists of air shipments between the two plants, and the slower or normal mode may represent all surface modes (e.g., ocean–local truck drayage). Finally, no shortages are allowed at both plants.

Since shipment and/or manufacturing lead times between plants could be significant, quantities of intermediate product families must accumulate at A before each delivery, and also perhaps at B to ensure the availability of intermediate products at B. Inventory carrying costs are incurred by products that are produced at A but unshipped, and those that have been shipped but are still in transit and have not arrived at B. Quantities of intermediate products received at B and unused during the period it arrives also incur inventory carrying costs.

Because of the product accumulation between inventory deliveries at plant A, the production sequence of product families on every individual production line would certainly affect A’s delivery schedule. Further, the production schedule of finished products at plant B could also potentially affect the intermediate product delivery frequency and choice of transport mode.

The relationships between the production and transportation decision outcomes suggest that the process industry described above could potentially benefit by integrating and coordinating its scheduling decisions. On the surface, it seems likely that a sequential approach to scheduling production and transportation could increase the system-wide cost significantly. For example, since A has a finite production capacity, it may be necessary to build and maintain larger inventories of the intermediate products to supply B. In addition, to avert any shortages of intermediate products at B, A may resort to utilizing the expedited transport mode more frequently, and thus incur a higher total transportation cost. However, because of the large number of interacting decision variables and parameters, the conditions under which coordinating efforts are most beneficial for the process industry remains to be found. In other words, the coordinated scheduling approach may not always be more attractive than the sequential scheduling approach if the benefits of the coordinated approach (such as cost savings) is insufficient to compensate for the increased complexity of solving to optimality an integrated scheduling problem.

3. Literature

The problem of simultaneously addressing the characteristics and requirements of the production and transportation functions to perform an overall optimization has attracted attention in recent years and some models have been proposed in this direction. Chandra and Fisher (1994) published a seminal paper on an integrated production, inventory, and distribution problem that investigates a two level system involving a single manufacturing plant producing finished goods, and routing the delivery of those goods to satisfy the requirements of several retailers. They show that an integrated analysis of the production scheduling and routing problems can yield a 3–20% reduction in operational cost. Interestingly, their computational results suggest that an integrated analysis reaps the most benefit in reducing cost only when more options become available which will improve the distribution schedule through consolidation. For example, the opportunity for consolidation increases when the planning horizon is long, and when many more products are produced and delivered to a larger number of customers. Also, the benefits of coordination are more pronounced as the ratio of the distribution costs to the production costs increases.

Research papers on three echelon systems with either a wholesaler or multiple warehouses with intermediary roles linking the manufacturer with retailers are as follows. Haq et al. (1991) study an integrated

production and distribution system where the manufacturer produces and ships batches of products to warehouses and from warehouses to retailers. The total inventory holding costs at all echelons, fixed distribution costs, and backlogging costs at the retailers is minimized. Ishii et al. (1988) consider a manufacturer, wholesaler, and retailers where products with short life cycles are produced and delivered via a pull system. The objective is to minimize the cost of obsolete products at the retailer.

Hahm and Yano (1995) analyze a system with a single supplier that produces multiple components on a single production line. The components, using a single transportation mode, are delivered directly to an assembly facility which uses them at a constant rate. No inventory is allowed to accumulate at the assembly facility and setup cost at the assembly facility is ignored.

In a recent paper that surveys the integrated analysis of production and distribution systems over the last two decades, Sarmiento (1999) cites the need to consider in the analysis the presence of common carriers to perform regular or emergency shipments. Most existing studies assume a single transportation mode with a fixed lead time. Sarmiento also notes the possible mismatch of time units for the production and transportation decisions as an important issue that requires attention. Our study meets these needs, i.e., we capture deterministic variations in the transportation time by the presence of carriers to perform either normal or expedited shipments. We also address the impact of potential differences in the definition of a production period and a transportation period in integrating the production and transportation scheduling decisions. For instance, a production period is a week with either 5, 6 or 7 working days while the intermediate product deliveries could be daily or only during certain days of a week. To our knowledge, none of these features have been incorporated in any single unified production and transportation scheduling model.

In this paper, we study the costs and benefits of coordinating the production and inter-facility transportation scheduling decisions of a process industry firm. We also aim to develop a better understanding of the general relationships between the production and transportation scheduling decisions, and how the changes in plant capacity, costs, and production specifications could affect the scheduling decisions as well as the choice of carriers to perform slow and fast shipments.

We present a mixed integer programming model formulation that evaluates numerous production and transportation costs and decision variables, plant capacities, and service tradeoffs which a process industry firm ideally must consider in developing its production schedules. The costs include setup, product changeover, intermediate product and finished goods inventory carrying costs, and transportation and in transit inventory carrying costs. The service tradeoffs include balancing long production runs which minimizes changeover requirements versus inventory carrying costs, and balancing the lower transportation costs associated with slower transport modes versus the resulting increased inventory investment requirements. Capacity considerations may include production possibilities and constraints at each plant, and perhaps limits on the available transportation capacity.

We developed strong valid linear inequalities and used them to strengthen the linear programming (LP) bound of our model. As a result, we were able to find near optimal production and transportation schedules with a standard LP-based branch-and-bound procedure. With real and simulated data from a pharmaceutical firm, we determined the cost advantage of coordinated scheduling. Our computational study shows that coordinated schedules can yield as much as 28% savings on total production and transportation costs over those obtained using sequential scheduling. We also tried empirically to identify other situations for which an integrated analysis would yield some cost savings.

4. Model formulation

Our model formulation of the integrated production and transportation scheduling problem coordinates three major decisions, i.e., the production scheduling decisions at plant A, the transportation scheduling

decisions in shipping intermediate products from A to B, and the production scheduling decisions at B. Our assumptions are as follows.

We view every production line at each individual plant as a single stage process capable of producing several product families. Also, we assume in this study that every product family has a designated production line. This is a reasonable assumption because the individual assignments of product families to specific production lines (or the loading problem) can be a separate decision which typically is based on technical and economic factors.

At any given time, a production line is either *shutdown*, idle but ready to produce (i.e., line is *idling*), or the line is *producing*. A production line is shutdown if it is not setup to produce any product. In our model formulation, setting up a line to produce product zero (a dummy product) means the line is shutdown. In contrast, a production line is idle if it is setup, but no units are produced. Whenever the production line is idling or producing, production personnel maintain the line (e.g., monitor and adjust the machines and check the product quality). Their salaries comprise the setup cost, i.e., the fixed cost of idling or producing on a line.

At the beginning of a production period, a changeover cost is incurred (e.g., cleaning cost, machine adjustment/fine tuning cost) in addition to the setup cost when the mode of a production line to produce one product changes to a different product. A production changeover also consumes time. In a number of process industries (see Miller, 2002), production changeovers occur during breaks or during times when no production is scheduled (e.g., grave yard shift or weekend). This avoids interfering with the regular production. Thus, it is reasonable in many cases to assume no production losses during a changeover. The basic changeover cost in these situations is the overtime wages of the crew performing the setup. It is important to note however that when a production week is 7 days long and the production lines operate 24 hours daily with no breaks (e.g., petro-chemical manufacturing), a product changeover may incur a production loss. Since our purpose in this paper is to study the benefits of coordination, considering the production loss unduly complicates our model while only marginally adding to our insights about coordination. Thus, we do not explicitly address cases where there are production losses resulting from changeovers in this model.

The planning horizon is divided into equally spaced production periods. The number of production days (which is also the number of business days) in each period is typically 5, 6 or 7 days long. The demand for finished goods in each production period is known. Also known are the starting inventories at plants A and B and those that are in transit.

The production line status (i.e., the product setup on the production line) at the beginning of period 1 is given. Because the production rates are fixed and products have designated production line assignments, the number of production weeks for each finished product and intermediate product is known. Ergo, the total manufacturing cost (which is the sum of direct material, labor and overhead costs) is a fixed cost, and thus we exclude it from our model.

The transportation or delivery lead time between A and B can range from a few hours to several weeks. Further, shipments may depart from A and arrive at B at the beginning, the end, or the middle of a production period. Typically, based on cost tradeoffs (e.g., freight costs and inventory carrying costs) and other considerations, a firm determines the alternative modes for transporting the intermediate product inventories from A to B. We assume that the transport modes (i.e., air, land and ocean transports) have unlimited capacities.

The production scheduling decisions at A and B, i.e., decisions on product setups, changeovers, production and inventory levels, are similar. To have a compact model formulation, we employ with clarity a single common set of notation to index the products and production lines and define the production variables and parameters of plants A and B. We distinguish certain notation individually when necessary to avoid ambiguity by using the index A or B to identify the particular plant. The details about our model are as follows. Denote by

k	index for plants,
i	index for products (utilized in the model to represent an intermediate product or a finished product),
j	alternative index for finished products,
ℓ	index for production lines,
t	index for time periods,
m	index for transportation mode,
K	the index set of plants, $K = \{A, B\}$,
I_k	the index set of products at plant k , $k \in K$. A product is an intermediate product if $i \in I_A$ and a finished product if $i \in I_B$. Note that $I_A \cap I_B$ is empty,
J_i	the index set of finished products which utilize the intermediate product $i \in I_A$ as the raw material input,
L_k	the index set of production lines at plant k . Note that $L_A \cap L_B$ is empty,
Q_ℓ	the index set of products production line ℓ can produce. Note $Q_\ell \cap Q_{\ell'}$ is empty for all ℓ and ℓ' in L_k , $k \in K$. Also, Q_ℓ includes product zero (which implies that line ℓ is shutdown) for all $\ell \in L_k$, $k \in K$,
T	the planning horizon,
p	the number of production days,
M	the index set of transport modes.

The notation used in the model to represent the different parameters are as follows.

$q_{i\ell}$	the production rate of product i ($i \in I_k$, $\ell \in L_k$, $k \in K$),
$f_{i\ell}$	the setup cost of product i on line ℓ ,
$s_{i\ell}$	the changeover cost of product i on line ℓ ,
h_{ik}	the unit carrying cost per period of intermediate product $i \in I_A$ at plant $k \in K$,
h'_{iB}	the unit carrying cost per period of finished product $i \in I_B$ at plant B,
d_{it}	the demand of finished product $i \in I_B$ in period t ,
u_{ij}	the amount of intermediate product $i \in I_A$ required to produce a unit of finished product $j \in I_B$,
e'_m	the lead time of transport mode m in days. Note that $e'_1 < e'_2 < \dots < e'_M$,
$e(m)$	the lead time of transport mode m expressed in terms of the number of production periods, i.e., $\lfloor \frac{e'_m}{p} \rfloor$ where $\lfloor \cdot \rfloor$ is the largest integer less than or equal to $\frac{e'_m}{p}$,
r_m	the fraction of shipment departing from A during period $t - e(m) - 1$ via transport mode m that arrives and can be utilized at B in period t . As will be explained later $r_m = \frac{e'_m}{p} - e(m)$,
$1 - r_m$	the fraction of shipment departing from A during period $t - e(m)$ via transport mode m that arrives and can be utilized at B in period t ,
a_{im}	the unit cost to transport and carry (in transit) product $i \in I_A$ by transport mode m .

The decision variables in our model formulation are given below.

$X_{i\ell t}$	$\begin{cases} 1, & \text{if on line } \ell \text{ product } i \text{ } (i \in Q_\ell, \ell \in L_k, k \in K) \text{ is produced in period } t; \\ 0, & \text{otherwise.} \end{cases}$
$U_{i\ell t}$	$\begin{cases} 1, & \text{if on line } \ell \text{ product } i \text{ is setup in period } t; \\ 0, & \text{otherwise.} \end{cases}$
$Z_{i\ell t}$	$\begin{cases} 1, & \text{if on line } \ell \text{ a product changeover to product } i \text{ occurs in period } t; \\ 0, & \text{otherwise.} \end{cases}$
W_{ikt}	the ending inventory of intermediate product $i \in I_A$ at plant $k \in K$ in period t ,
W'_{iBt}	the ending inventory of finished product $i \in I_B$ in period t ,

Y_{imt} the quantity of intermediate product $i \in I_A$ shipped in t by transport mode m .

Our model formulation is as follows.

Problem(P)

$$\begin{aligned} \text{Minimize} \quad & \sum_{k \in K} \sum_{\ell \in L_k} \sum_{i \in Q_\ell} \sum_{t=1}^T (f_{i\ell} U_{i\ell t} + s_{i\ell} Z_{i\ell t}) + \sum_{i \in I_A} \sum_{k \in K} \sum_{t=1}^T h_{ik} W_{ikt} \\ & + \sum_{i \in I_B} \sum_{t=1}^T h'_{iB} W'_{iBt} + \sum_{i \in I_A} \sum_{m \in M} \sum_{t=1}^T a_{im} Y_{imt} \end{aligned} \quad (1)$$

subject to:

$$\sum_{i \in Q_\ell} U_{i\ell t} = 1, \quad \ell \in L_k, \quad k \in K, \quad t = 1, \dots, T, \quad (2)$$

$$0 \leq X_{i\ell t} \leq U_{i\ell t}, \quad \text{integer}, \quad i \in Q_\ell, \quad \ell \in L_k, \quad k \in K, \quad t = 1, \dots, T, \quad (3)$$

$$Z_{i\ell t} \geq U_{i\ell t} - U_{i\ell t-1}, \quad i \in Q_\ell, \quad \ell \in L_k, \quad k \in K, \quad t = 1, \dots, T, \quad (4)$$

$$W'_{iBt} = W'_{iBt-1} + q_{i\ell} X_{i\ell t} - d_{it}, \quad i \in Q_\ell, \quad \ell \in L_B, \quad t = 1, \dots, T, \quad (5)$$

$$W_{iAt} = W_{iAt-1} + q_{i\ell} X_{i\ell t} - \sum_{m \in M} Y_{imt}, \quad i \in Q_\ell, \quad \ell \in L_A, \quad t = 1, \dots, T, \quad (6)$$

$$\begin{aligned} W_{iBt} = W_{iBt-1} + \sum_{m \in M} (r_m Y_{imt-e(m)-1} + (1-r_m) Y_{imt-e(m)}) \\ - \sum_{\ell \in L_B} \sum_{j \in (J_i \cap Q_\ell)} u_{ij} q_{j\ell} X_{j\ell t}, \quad i \in I_A, \quad t = 1, \dots, T, \end{aligned} \quad (7)$$

$$0 \leq U_{i\ell t} \leq 1, \quad \text{integer}, \quad i \in Q_\ell, \quad \ell \in L_k, \quad k \in K, \quad t = 1, \dots, T, \quad (8)$$

$$Z_{i\ell t} \geq 0, \quad i \in Q_\ell, \quad \ell \in L_k, \quad k \in K, \quad t = 1, \dots, T, \quad (9)$$

$$Y_{imt} \geq 0, \quad i \in I_A, \quad m \in M, \quad t = 1, \dots, T, \quad (10)$$

$$W_{ikt} \geq 0, \quad i \in I_k, \quad k \in K, \quad t = 1, \dots, T, \quad (11)$$

$$W'_{iBt} \geq 0, \quad i \in I_B, \quad t = 1, \dots, T. \quad (12)$$

Our objective is to minimize the total of the setup and changeover costs at plants A and B, the intermediate product inventory cost at A and B, the cost of finished good inventory at B, and the cost of transporting intermediate products from A to B. Note that the cost of carrying a unit of an intermediate product $i \in I_A$ at plant A (i.e., h_{iA}) may differ from that at plant B (h_{iB}). The constraints are as follows. Only one product can be produced on each line in every period (constraints (2)). Production is possible only when the production line is setup (constraints (3)). If the production line is setup to produce a real product and the production quantity is zero (e.g., $U_{i\ell t} = 1$ and $X_{i\ell t} = 0$), then the production line is construed idle in period t . A changeover on a production line to a specific product in period t occurs only if the line is setup for that product in period t and for a different product in period $t-1$ (constraints (4)).

The weekly demands of finished products must be satisfied (constraints (5)). Constraints (6) and (7) are flow conservation constraints of intermediate products between plants A and B. Constraints (7) stipulate that in any given period, plant B is able to produce finished goods only if it has the necessary intermediate products that it needs for production in inventory. Constraints (7) allow B to receive and process into finished goods small batch shipments of intermediate products that can depart from A daily or on certain days during a period t_0 and perhaps also in periods $t < t_0$. Because of the shipment lead time, the aggregate daily shipments received by B during a period $t \geq t_0$ often only constitute certain fractions (represented by r_m and $1-r_m$) of the total daily shipments from A during periods $t \leq t_0$. One can best understand the impact of constraints (7) and our interpretation of r_m by examining the example below.

Let the transportation lead time from A to B be 2 days (i.e., $e'_1 = 2$). Assume a p of 5 business days. Thus, $e(1)$ is 0. Assume for simplicity that the shipment of any intermediate product i from A to B via a transport mode m in period t of quantity Y_{imt} is delivered in p equal daily shipments of uniform batches each of size $\frac{Y_{imt}}{p}$. For instance, if $Y_{imt} = 1000$, then 200 units would leave A for B daily. The timing of shipment arrivals at B would depend on the transport mode. For example, $\frac{3}{5}$ or 600 units of the total quantity of intermediate product shipped from A to B during production period t would arrive and be available for processing at B in period t when the transit time is 2 days. Specifically, 200 units depart A on Monday of period t and arrives at B on Tuesday and will be available for production at B on Wednesday also in period t . Another 200 units depart A on Tuesday and 200 units on Wednesday and are available for production at B on Thursday and Friday respectively also in period t . The remaining $\frac{2}{5}$ or 400 units would not be available for finished good production at B until period $t + 1$, i.e., shipments of 200 units each depart A on Thursday and Friday and arrive at B on Monday and Tuesday respectively in period $t + 1$. Using the same argument, $\frac{2}{5}$ of the total shipment in period $t - 1$ (if any) arrives at B and are used for finished goods production in period t . Note here we are assuming no production nor shipment take place at A and B on Saturday and Sunday of each week. We can easily capture the cost of holding inventory in transit during Saturday and Sunday in the unit transportation cost a_{im} . Finally, in this example $r_1 = \frac{3}{5}$ and $1 - r_1 = \frac{2}{5}$, and hence the general expression for r_m is $r_m = \frac{e'_m}{p} - e(m)$. We shall use this expression for the remainder of the paper.

To conclude this section, our interpretation of r_m and constraints (7) as well would make sense under two mutually exclusive shipping strategies given below.

- Uniform shipments—transporting the quantity $Y_{imt} > 0$ in p equal daily shipments.
- Nonuniform shipments—shipping lumped quantities on fixed designated days. Using the preceding example, instead of the Monday and Tuesday shipments of 200 units each departing from A during period t , a single consolidated shipment of 400 units would leave A during period t on Tuesday morning which would arrive at B on Thursday of period t and be available on Friday for processing at B. Further, the 600 units would leave A in period t on Friday and arrive at B on Monday and be available for processing at B on Tuesday of period $t + 1$.

Either shipping policy is reasonable. In some industries and firms, more frequent deliveries of relatively smaller shipments are preferred as firms try to trim down inventory and enhance production flexibility. However, other firms choose a more consolidated shipping approach with lumped shipment quantities departing A on certain fixed days. Finally, r_m could be based on some other measures that will suit the needs and match the available resources of a firm.

5. Solution approach

Because of the large number of integer variables for even a small problem, obtaining integer solutions for (P) via a branch-and-bound procedure requires evaluating a large number of branches and performing a large number of simplex pivots. To reduce the time required to find a low cost integer solution, we add strong valid inequalities to our problem. The valid inequalities exploit the underlying structure of the problem which serve to increase the optimal value of the LP relaxation of the problem. It also tend to drive more of the relaxed integer variables to assume binary values. We shall present a modified formulation of problem (P) and then show how we generate the valid inequalities.

We modify problem (P) by eliminating every inventory variable (i.e., W_{ikt} and W'_{iBt}) in the model formulation. In particular, we replace constraints (5), (6), (7), (11), and (12) with restrictions consisting only of production variables (X) and shipment variables (Y) as shown below.

$$\sum_{n=1}^t q_{i\ell} X_{i\ell n} \geq \sum_{n=1}^t d_{in} - W'_{iB0}, \quad i \in Q_\ell, \quad \ell \in L_B, \quad t = 1, \dots, T, \quad (5')$$

$$\sum_{n=1}^t q_{i\ell} X_{i\ell n} \geq W_{iA0} + \sum_{m \in M} \sum_{n=1}^t Y_{imn}, \quad i \in Q_\ell, \quad \ell \in L_A, \quad t = 1, \dots, T, \quad (6')$$

$$\sum_{n=1}^t \sum_{m \in M} ((1 - r_m) Y_{imn-e(m)} + r_m Y_{imn-e(m)-1}) \geq \sum_{n=1}^t \sum_{\ell \in L_B} \sum_{j \in (J_i \cap Q_\ell)} u_{ij} q_{j\ell} X_{j\ell n} - W_{iB0} - Y_{i0}, \quad i \in I_A, \quad t = 1, \dots, T. \quad (7')$$

Note that W_{iA0} , W'_{iB0} , W_{iB0} , and Y_{i0} are constants representing the starting product inventories at plants A and B and the in transit inventories of intermediate products prior to period $t = 1$. Making similar changes in the objective function and combining like terms together, problem (P) is now written as

Problem (P₁)

$$\begin{aligned} \text{minimize} \quad & \sum_{t=1}^T \sum_{k \in K} \sum_{\ell \in L_k} \sum_{i \in Q_\ell} (f_{i\ell} U_{i\ell t} + s_{i\ell} Z_{i\ell t}) + \sum_{i \in I_A} \sum_{\ell \in L_A} \sum_{t=1}^T c'_{i\ell t} X_{i\ell t} \\ & + \sum_{j \in I_B} \sum_{\ell \in L_B} \sum_{t=1}^T c''_{j\ell t} X_{j\ell t} + \sum_{i \in I_A} \sum_{m \in M} \sum_{t=1}^T a'_{imt} Y_{imt} + R \end{aligned} \quad (1')$$

subject to: (2), (3), (4), (5'), (6'), (7'), (8)–(10).

The objective function coefficients in (P₁) are

$$c'_{i\ell t} = q_{i\ell} h_{iA} (T - t + 1) \quad \text{for } i \in Q_\ell, \quad \ell \in L_A, \quad t = 1, \dots, T,$$

$$c''_{j\ell t} = q_{j\ell} \left(h'_{jB} - \sum_{i: j \in J_i} h_{iB} u_{ij} \right) (T - t + 1) \quad \text{for } j \in I_B, \quad \ell \in L_B, \quad t = 1, \dots, T,$$

$$a'_{imt} = a_{im} + (r - h_{iA}) (T - t + 1) - r \frac{e'_m}{p} \quad \text{such that for } i \in I_A,$$

$$r = h_{iB} \quad \text{for } 1 \leq t \leq T - e(m) \quad \text{and}$$

$$r = 0 \quad \text{for } T - e(m) + 1 \leq t \leq T,$$

$$R = \sum_{i \in I_A} T(h_{iA} W_{iA0} + h_{iB} W_{iB0}) + \sum_{j \in I_B} \sum_{t=1}^T h_{jB} \left(W'_{jB0} - \sum_{n=1}^t d_{jn} \right) + \sum_{i \in I_A} \sum_{m \in M} \sum_{t=1}^{e(m)} h_{iB} (T - t + 1) Y_{imt-e(m)}.$$

Note R is a constant and thus can be deleted from the objective function of problem P₁.

Since no shortages of finished products are allowed at B, we can generate and add to our model valid linear constraints on the cumulative quantities of the intermediate products that must be produced in each period at plant A by offsetting the cumulative production of the intermediate product by the lead time of the fastest transport mode. We explain this with a simple example. Assume that a unit of a finished product requires a single unit of an intermediate product. Let the net cumulative demand of a finished product (i.e., the cumulative demand less the initial finished goods inventory of the product) at the beginning of some period t be q units. Also, let us assume that there is no transportation lead time to ship a unit of the intermediate product from A to B. In order for a production schedule to be feasible, we must have at the

beginning of period t a net cumulative production of the intermediate product of at least q units. However, if the transportation lead time was $e > 0$, then the net cumulative production of the intermediate product must be at least q units at the beginning of $t - e$ periods.

We now determine the appropriate value for e . Using the longest transport lead time for e (which is tantamount to transporting the intermediate product via the slowest mode) would automatically create unnecessary inventory of the intermediate product because it forces us to build the intermediate product to satisfy q units early. This is avoided however if we use the shortest transport lead time for e . Additionally, our model still retains its flexibility to satisfy the net cumulative production requirement of the intermediate product in periods earlier than $t - e$. Therefore, the general expression for the valid inequality is as follows.

$$\sum_{n=1}^{t-e} q_i X_{i\ell n} \geq \sum_{j \in J_i} \sum_{n=1}^t u_{ij} d_{jn} - W_{iA0} - W_{iB0} - Y_{i0}, \quad i \in Q_\ell, \ell \in L_A, t = 1, \dots, T. \quad (13)$$

Denote by (P_2) problem (P_1) with constraints (13) added. We note that the optimal solutions to problems (P) , (P_1) , and (P_2) are identical.

Problem (P_2) with the exclusion of constraints (2), $(6')$ and $(7')$ decomposes into several independent single product, single production line subproblems for plant A and plant B. Each subproblem can be represented as a fixed charge network flow model. We can generate the valid inequalities from each of these network flow models and append them to the LP relaxation of problem (P_2) , i.e., problem (P_2) with the integrality restrictions on constraints (3) and (8) relaxed. The valid inequalities are facets originally developed by Magnanti and Vachani (1990) for the single product, single production line problem. Using the valid inequalities derived from the network flow models to strengthen the LP bound of (P_2) , we can then solve (P_2) directly by conventional branch-and-bound techniques.

We now show how to generate the valid inequalities. Consider the acyclic fixed charge network flow graph shown in Fig. 1. It represents a single product, single production line scheduling problem with three periods and two units of demand. The network has 10 nodes. The six nodes in the center, which are labeled with the production period t'' or t' , are transshipment nodes. The three nodes on the right are demand periods t with one unit of demand each at $t = 1$ and $t = 3$. The single node S on the left is a source node with two units of supply. The variable X_{it} is a flow variable which is one if the solution sends a flow from t' to t . This arc has a unit capacity and the cost of a unit flow on this arc is c'_{it} for an intermediate product and c''_{jlt} for a finished product. The flow on arc $(t, t + 1)$ is W_{ikt} , the ending inventory in period t , and the cost is zero since c'_{it} and c''_{jlt} already include the holding cost. The variables Z_{it} and U_{it} are indicator variables where each variable is equal to one if the solution sends a positive flow from S to node t'' and node t'' to node t' , respectively. Thus, for example, a flow on arc $(2'', 2')$ implies $U_{i2} = 1$. This could mean that $U_{i1} = 1$ or $Z_{i2} = 1$, i.e., sending a flow on arc $(1'', 1')$ or $(S, 2'')$. The cut is obtained, for example, by a cutset (Q, Q') with $Q' = \{3'', 2', 3', 2, 3\}$ and $Q = V \setminus Q'$ where V is the node set. Since the set Q' has one unit of demand that must be met by a flow from Q to Q' , the inequality $W_{ik1} + U_{i2} + Z_{i3} + 1 \geq 2$ is valid. But $W_{ik1} = X_{i1} - 1$, therefore $X_{i1} + U_{i2} + Z_{i3} \geq 2$. Notice that variable Z may exceed one when we append this

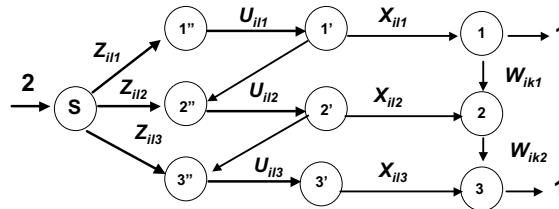


Fig. 1. The network flow diagram for generating valid inequalities.

inequality to the LP relaxation of (P_2) . Thus, we need to add to (P_2) the constraints $Z_{ilt} \leq 1, i \in Q_t, \ell \in L_k, k \in K, t = 1, \dots, T$.

We coded our procedure that generates the valid inequalities in C and used CPLEX 6.5's integer programming optimizer (CPLEX 6.5 Reference Manual, 1999) to solve numerous problems on a desktop computer with a 400 megahertz Pentium II processor. The next section provides the details of our computational study and reports our results.

6. Computational study

Under different production and transportation environments and cost structures, we determine in this section the potential cost savings that can accrue from coordinated scheduling, and study some of the key features of our model solutions utilizing a set of actual demand data and cost parameters of a pharmaceutical manufacturing firm.¹ A brief description of the firm is as follows.

The firm produces two major product families, powder (e.g., aspirin) and solution (e.g., cough syrup). Of particular interest in this study are the powder plants A and B that produce the intermediate and finished products, respectively. Plants A and B each have one dedicated production line to produce and package, respectively, six powder product families. The production bill of materials requires a single intermediate product for every unit of finished product. Production lines at A and B operate 5 days a week, but A runs three shifts daily and B only two shifts daily. Plant A produces batches of chemical compounds (intermediate products) through a series of unit operations such as drying, crystallization, and extraction. These batches are shipped to B which processes them through a combination of mixing, granulating, filling, and packaging steps into tablets, capsules, and bottles of consumer ready drugs.

The firm currently uses the sequential scheduling approach to schedule the production at plants A and B and the inter facility transfer of intermediate products from A to B. Using a computer spreadsheet software to develop feasible schedules, the quality of the schedules relies solely on the scheduler's experience. In this study, we simulate the firm's production and transportation scheduling environment and compare the sequential schedules that are generated with our model solutions. The details of the study are as follows.

6.1. The design of our computational study

The purpose of this study is to test how the key features of our model solutions, e.g., the intermediate product versus finished good inventory levels and production and transportation schedules, might vary under different production environments, cost parameters, product demands, and transportation lead times. The study consists of seven *independent* cases. Under each case, we vary the product demands and production and transportation lead times. The details are as follows.

Case I is our base case which represents the firm's *actual* production and transportation environment and cost structure. Cases II–VII are variations of Case I and they are described as follows. Case II adds two more production lines at A and assigns two product families to each line. Case III sets the intermediate product carrying cost rates at B to three times that of A's. Case IV uses unit transportation costs that are three times larger than those of the base case. Case V assumes a bill of materials which requires three distinct intermediate products for each unit of a finished product.

Cases VI and VII use cost parameters that are significantly different than those of Cases I to V. In particular, Case VI's intermediate product and finished goods carrying costs were respectively seven and

¹ The pharmaceutical firm referenced here is not the firm where the second author currently works. It is an unnamed firm for whom the first author has consulted.

three times less than that of Case I. The setup costs and changeover costs in Case VI were three times larger than those in Case I. Finally, Case VI's expedited and normal transport modes were seven times less than Case I's. Case VII's cost parameters are identical with Case VI's except the expedited and normal transportation costs are two times and three times that of Case I's, respectively.

For each of the seven cases, we vary the product demand pattern, production lead time and transportation lead times. Specifically, we consider three 26 period finished product demand data sets, namely, a set of actual demand data obtained from the firm, and two sets of simulated demand data with increasing and lumpy sales patterns. The demand data with increasing sales are the actual demands sorted in ascending order. To generate lumpy demands from the actual demand data, for each product family, we randomly picked two or three consecutive periods, e.g., t , $t + 1$, and $t + 2$ and moved the demands in periods, t and $t + 1$ to period $t + 2$. For each sales pattern, we varied the length of an individual production period by setting it to either 5 or 7 days. Finally, for each of the two different production periods, the expedited and normal transportation lead times were 1 day and 5 days or 2 days and 10 days, respectively. Thus, under each of the seven cases, we solved 12 different problems representing the different production environments (i.e., three finished goods sales patterns, two production lead times and two sets of transportation lead times).

For every problem that we solved, we were unable to assess the actual cost savings of a coordinated schedule to the firm because the actual schedule implemented for Case I was unavailable. Instead, we generated simulated sequential schedules by first finding an optimal production schedule for plant B (i.e. we took the production scheduling part of our model that pertained to plant B and solved it to optimality). Using plant B's schedule, we computed the intermediate product requirements and determined the optimal production and transportation schedules for plant A (i.e. we fixed all the decision variables in our model corresponding to plant B and then optimized our model to obtain the production and transportation schedules for plant A). Note that the sequential schedules we obtained could be far better than the actual schedules because the complexity of scheduling production and transportation manually precludes any attempt to obtain optimal solutions.

To ensure feasibility, we introduced in every problem modest amounts of starting intermediate product inventories. For each case, the minimum feasible inventory quantities were obtained by solving every problem variation in that case several consecutive times using the sequential scheduling approach presented above. Beginning with 4 weeks worth of production as starting inventories, we gradually reduced the quantities each time by a half week and solved the problem until the problem becomes infeasible. The largest of feasible inventory quantities found among the 12 problem variations in each case was used in every problem belonging to that case. We note that the smallest feasible starting inventory for a sequential schedule is also feasible for the corresponding coordinated schedule, but not vice versa.

Finally, for every problem, we determined the best coordinated schedules and sequential schedules. When running each individual test problem, we stopped either when the integrality gap, i.e., the difference between the best integer objective value and the objective value over the best integer value of the best node remaining was at most 1.5% or the CPU time reached 1800 seconds. The discussion of our results are as follows.

6.2. *Presentation and analysis of results*

We grouped our results according to demand pattern, production lead time and transportation lead times, and reported the minimum, average and maximum savings per case in Table 1. We computed the savings as the difference between the cost of the sequential schedule less the cost of the coordinated schedule over the cost of the sequential schedule.

Highlighting the important outcomes of this study, Case I which represents the actual problem yielded an overall savings of 5.9% with a minimum and maximum savings of 2.5% and 11.8%, respectively. Case IV

Table 1

Percentage cost savings between coordinated and sequential schedules for Cases I, II, III, IV, V, VI and VII

Problem variation	No. of problems per case	Case ^a																				
		I			II			III			IV			V			VI			VII		
		Min	Ave	Max	Min	Ave	Max	Min	Ave	Max	Min	Ave	Max	Min	Ave	Max	Min	Ave	Max	Min	Ave	Max
<i>Demand pattern</i>																						
Actual	4	3.5	6.0	7.2	0.9	2.7	5.0	8.2	10.3	13.5	1.5	18.5	28.1	5.1 ^a	8.2	9.9	0.4	5.5	10.8	0.7	4.1	7.1
Increasing	4	2.5	4.5	5.9	2.0 ^a	3.1	4.0	5.7	8.6	11.7	6.2	19.0	27.2	4.4 ^a	4.4	4.4	1.6	6.1	9.4	2.5	4.8	7.3
Lumpy	4	3.0	7.2	11.8	2.5	4.8	8.8	6.1	7.4	9.3	7.3	15.8	18.7	9.8 ^a	13.4	18.2	1.6	5.1	9.3	3.8	8.3	13.3
<i>Production lead time</i>																						
5 days	6	4.2	7.2	11.8	3.4	4.9	8.8	6.1	9.3	13.5	17.1	21.5	28.1	4.4 ^a	7.9	12.2	3.6	6.2	9.3	3.3	6.8	13.3
7 days	6	2.5	4.7	7.1	0.9 ^a	2.1	4.0	5.7	8.3	11.8	1.5	14.1	25.9	9.7 ^a	12.6	18.2	0.4	5.0	10.8	0.7	4.7	11.1
<i>Transportation lead times</i>																						
1 and 5 days	6	2.5	4.8	6.9	1.0 ^a	3.2	4.2	5.7	6.8	8.2	1.5	11.6	18.7	5.1 ^a	7.5	9.8	0.4	2.6	4.5	0.7	3.3	5.1
2 and 10 days	6	4.2	7.0	11.8	0.9	3.9	8.8	7.7	10.7	13.5	18.0	23.9	28.1	4.4 ^a	10.9	18.2	5.9	8.5	10.8	5.0	8.1	13.3
Overall percentage cost difference		2.5	5.9	11.8	0.9	3.6	8.8	5.7	8.8	13.5	1.5	17.8	28.1	4.4	8.7	18.2	0.4	5.6	10.8	0.7	5.7	13.3

Case descriptions: Case I: Base case (i.e., one production line at each plant, one intermediate product required for every finished product, and original cost parameters from the pharmaceutical firm).

Case II: Larger production capacity at plant A (three production lines at plant A and one production line at plant B).

Case III: Carrying cost rates of the intermediate products at plant B are three times that at plant A.

Case IV: Transportation costs are three times that of Case I.

Case V: The bill of materials is 3 distinct intermediate products required for every finished product.

Case VI: The intermediate product and finished goods carrying costs were seven and three times less than that of Case I. Its setup and changeover costs were three times larger than Case I, and expedited and normal transport modes were individually seven times less than Case I's.

Case VII: The same cost parameters as Case VI except the expedited and normal transport modes were two times and three times more than Case I's, respectively.

Note that these are independent cases.

^aThe minimum, average and maximum % cost savings reported under Case II excluded one problem with at least 10% integrality gap from consideration. The minimum, average and maximum % cost savings reported under Case V excluded 1, 3, 1, 2, 3, 4, and 1 problem from the 7 respective problem variations with at least 10% integrality gap from consideration. Running CPLEX a total of 2 cpu hours failed to reduce the integrality gaps to less than 5%.

results produced the highest savings, revealing that the firm could save on average 17.8% from coordinated schedules under a scenario where the transportation costs are three times higher than those in Case I. When the transportation lead times were 2 days and 10 days for Case IV, the average savings could reach as high as 23.9%. Note that as the differential in transport lead times increased (e.g., from 1 day lead time for expedited shipment and 5 days for normal shipment to 2 days and 10 days, respectively), the volume of expedited shipments also increased significantly in sequential schedules because some intermediate products, after production at plant A, must be transported quickly to plant B in order for plant B to meet its production schedule for finished products. (Recall that in sequential scheduling, the finished product production schedule at B *drives* the intermediate product production and transportation schedules at A.)

Interestingly, Case III, which eliminated 98% of intermediate product inventory at B in favor of making deliveries to B only if B needs them, yielded an overall average savings of 8.8%. We provide more comments about Case III later in this section. Cases VI and VII also yielded significant savings for coordinated schedules, on average 5.6% and 5.7%, respectively.

In Table 1, we note that the minimum, average and maximum percentage cost savings reported under Case II excluded one problem with an integrality gap exceeding 10% from consideration. The minimum, average and maximum percentage cost savings reported under Case V exclude 1, 3, 1, 2, 3, 4, and 1 problems from the seven respective problem variations because of integrality gaps exceeding 10%. Running CPLEX a total of 2 cpu hours failed to reduce the integrality gaps to less than 5%.

A review of modeling results across all cases reveals at least one intuitive pattern. On average, as network capacity or flexibility decreases (and the problem essentially becomes more tightly constrained), the potential savings produced by a coordinated approach—versus a sequential approach—appears to increase. Similarly, as certain production and transportation costs increased, again the coordinated approach versus the sequential approach appears to yield relatively higher potential savings. Table 2 illustrates these observations.

Note in Table 2 that in Case II where capacity increases, the average potential savings yielded from coordinated schedules decreases to 3.6% (as compared to the 5.9% average of Case I problems). In Cases III through V, either costs increase or capacity decreases, and the average savings produced by coordinated schedules range from 8.7% to 17.8%—all greater than average savings in Case I. Finally, with a mix of increases and decreases in certain costs, Cases VI and VII exhibit about the same overall average savings potential (5.6% and 5.7%) as does Case I (5.9%) when a coordinated scheduling model is utilized.

Table 2
Comparison of overall average savings of all problems in each case

Case	Change in capacity or cost relative to Case I	Description of change in capacity or costs relative to Case I	Overall average savings	Change in average savings versus Case I
I	—	—	5.9%	—
II	Increase capacity	Add two production lines at plant A	3.6%	(2.3%)
III	Increase costs	Raise inventory carrying costs	8.8%	2.9%
IV	Increase costs	Raise transportation costs to three times Case I	17.8%	11.9%
V	Decrease capacity	Change bill of materials for finished goods from 1 intermediate good to 1 finished good to 3 to 1 (effectively reducing capacity)	8.7%	2.8%
VI	Mixed increase and decrease costs	Costs of some parameters increased, while others are lowered versus Case I	5.6%	(0.3%)
VII	Mixed increase and decrease costs	Costs of some parameters increased, while others are lowered versus Case I	5.7%	(0.2%)

Table 3
Percentage cost differences between coordinated and sequential schedules

Case	No. of problems	Cost component					
		Setup and changeover	Intermediate product inventory	Finished goods inventory	Expedited shipment	Normal shipment	Total
I	12	2.7%	25.4%	−25.7%	68.0%	−4.9%	5.9%
II	11 ^a	0.4	12.3	−18.9	63.8	−5.9	3.6
III	12	2.7	33.3	−28.8	70.8	−7.3	8.8
IV	12	2.0	23.8	−21.8	75.1	−6.2	17.8
V	7 ^a	2.0	−0.1	−48.8	71.6	−8.7	8.7
VI	12	−2.7	2.9	5.6	88.4	−7.2	5.6
VII	12	−2.0	−0.9	6.6	90.0	−7.2	5.7
Ave.		0.7	10.2	−15.2	65.8	−6.1	7.2

The percentage cost difference for each cost component is computed as (uncoordinated schedule cost − coordinated schedule cost)/uncoordinated schedule cost × 100%.

^a The percentage cost differences between coordinated and sequential schedules reported under Case II excluded one problem with at least 10% integrality gap from consideration. Also, for Case V we excluded 5 problems with at least 10% integrality gap from consideration. Running CPLEX a total of 2 cpu hours failed to reduce the integrality gaps to less than 5%.

Table 3 displays the individual percentage savings in setup and changeover, intermediate product inventory, finished goods inventory, expedited and normal transportation, and total costs when coordinated schedules were used. For example, in Case I, the average percentage difference in changeover and setup costs between the coordinated and sequential schedules was 2.7%. The average difference in intermediate product inventory carrying costs, finished goods inventory carrying costs, expedited and normal shipment costs were 25.4%, −25.7%, 68.0%, and −4.9%, respectively. We shall explain shortly why the setup and changeover costs are insignificant sources of savings.

Coordination significantly reduces the use of the expedited transportation mode as illustrated by the 66% lower average cost of the coordinated schedules versus the sequential schedules among Cases I through VII. This represents one clear pattern that occurred across all cases; namely, that the coordinated scheduling approach reduces the need for expedited transportation.

Another pattern observed was that coordination has a miniscule effect in reducing the number of product changeovers at both A and B. In almost every problem that we solved, the number of changeovers between coordinated and sequential schedules was almost identical. This is expected since every product family at A and B is assigned to a single production line and there is no cost incentive to produce the intermediate products in quantities larger than needed to satisfy the demands for finished goods. However, the timing of product changeovers differ significantly with coordination. That is, production of intermediate products at A are synchronized with the need for those products at B. Not only did the intermediate product inventory carrying cost decrease with coordination, but also larger amounts of intermediate products were shipped via the less expensive normal mode.

The percentage cost difference between coordinated and sequential schedules in Table 3 showed a decrease in the intermediate product inventory level and an increase in finished good inventory level because in sequential scheduling we minimize the total production and inventory cost of finished goods at plant B independent of the production costs at plant A and the transportation costs of shipping the intermediate products from A to B. In contrast, we minimize the *total system cost* in coordinated scheduling. Thus, in a sequential schedule since the intermediate product shipment quantities are derived from the requirements of a locally optimized production schedule for finished goods at plant B and are merely passed on as production and shipment requirements to plant A, it is likely that the intermediate product inventory costs of the sequential schedule to be larger than or equal that in the corresponding coordinated schedule. Since the intermediate product production, inventory and shipment decisions are integrated with the production

decisions at plant B in a coordinated scheduling approach, the cost tradeoffs and economies of scale can potentially decrease the intermediate product inventory cost and increase the finished good inventory cost.

It is also important to recall that these tests compare the solutions of two optimization model-based approaches: a coordinated modeling method versus a sequential, non-coordinated method. Many firms (including those that the disguised test data is based upon) still employ sequential, non-optimization based methods (e.g., spreadsheets) to generate individual plant productions schedules. Further, as noted earlier, production schedules of intermediate plants and finishing plants may not even be directly linked. In such cases, the potential savings that an integrated optimization methodology could produce could significantly exceed the potential savings observed in our coordinated optimization versus sequential optimization model-based tests.

Finally, these tests were conducted in an environment where the variable manufacturing costs of solutions generated by the two modeling methods were identical. We did not model an environment where multiple production lines at a plant or plants could produce the same product at different costs. In this more complex situation, optimization-based production scheduling models typically identify much higher potential savings because the models can exploit the line manufacturing cost differences, as well as other cost factors. Because variable manufacturing costs were constant in these tests (i.e., the coordinated and sequential models have identical variable production costs), the models could essentially only evaluate the tradeoffs among changeover/setup costs, freight costs and inventory carrying costs to determine the cost-minimizing feasible solution. (These are the results shown in Tables 1–3.)

7. Brief note on managerial implications

This production and transportation model highlights the importance of employing an integrated approach towards scheduling a multi-echelon manufacturing system. Unless one views the multi-echelon problem in an integrated methodology (i.e., unless one simultaneously considers two echelons and the transportation links between them), the likelihood exists that a firm will operate its network at efficiency levels significantly below potential. A single echelon and/or single functional planning and scheduling approach almost by definition leads to sub-optimal results. The coordinated versus sequential optimization modeling test results just reviewed illustrate this. Further, as noted, these (optimization versus optimization) results are unable to fully capture the magnitude of potential savings possible when a firm uses an integrated multi-echelon optimization model-based scheduling approach to replace or complement a non-optimization based sequential method.

Although multi-echelon scheduling problems such as the one formulated in this paper often still present a daunting challenge to algorithm and software developers and practitioners, more and more such problems can now be solved and implemented. For managers and practitioners, this suggests that a more proactive approach towards addressing multi-echelon scheduling problems with sophisticated decision support tools represents an important opportunity to consider.

Finally, even if a firm does not explicitly utilize the type of scheduling model proposed here, taking this problem into consideration “conceptually” when scheduling plants and transportation links will lead to improved decision-making. That is, managers should consider the tradeoffs among such costs as production, inventory and transportation across multiple echelons of a manufacturing network, even if they do not evaluate these tradeoffs directly in an integrated production and transportation scheduling model. Our experience in industry suggests that firms are often unable to weigh these tradeoffs across echelons as closely as they should. Instead an echelon-by-echelon, function-by-function scheduling approach often prevails, again leading to less efficient operational execution than necessary.

8. Conclusion

We developed in this paper a dynamic production and transportation decision model to simultaneously determine the cost minimizing quantities of products an intermediate plant must produce and ship to a finishing plant using different transportation modes. Simultaneously, we also find the cost minimizing quantities of products that the finishing plant must produce to meet its customer demand on time. One of several benefits of coordinating production and transportation decisions is that it helps control the use of the expedited transport mode to a minimum through timely shipments of sufficient intermediate product quantities via the normal inexpensive mode to meet the input requirements of the finishing plant. It also reduces the system wide intermediate product inventory holding cost.

Further research in this area could take into consideration different production possibilities for each individual product family, i.e., having several production lines with different production rates and costs able to produce a product family. In this case, the selection of a production line to be assigned to a product family is a decision variable. The sourcing decision becomes an important issue if the supplies of intermediate products originate from various sources. Adding these and many more features to our problem would certainly increase its complexity and efficient heuristic approaches are needed to solve them.

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